AN AMPLE FAMILY ON X FROM AN AMPLE FAMILY ON THE BASE AND AN F-AMPLE INVERTIBLE MODULE

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 23312-23335, immediately following the statement at 23302-23310.

Editorial note. One cross-chapter pointer (Properties) is recorded in assembly.external_references; the rest of the inputs are definitional citations inside the chapter.

Editorial note. Related card S9-morph-get-ample-family (L26019) treats the different statement on Cartier divisors; the two share no span.

Allowed standard prerequisites (editorial).

- Definition of an ample family of invertible modules. Morphisms, Definition morphisms-definition-family-ample-invertible-modules, cited at proof lines 23314-23319. Fixes what has to be produced on X: enough sections whose nonvanishing loci are affine and cover.
- Relative ampleness. Morphisms, Definition morphisms-definition-relatively-ample, cited at proof lines 23312-23314. f-ample gives quasi-compactness of f and the ample-over-preimage behaviour of L used in the local step.
- Extending sections after inverting a function. Properties, Lemma properties-lemma-invert-s-sections, cited at proof lines 23325-23327. Extends the local section s of L over X_{f^*t} to a global section s' of f^*N^{tensor n} tensor L after multiplying by (f^*t)^n.

Source reference. Original reference 0FXR: focused-seeds/geometry/S9-morph-ample-family-ample-relative/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01VH: focused-seeds/geometry/S9-morph-ample-family-ample-relative/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01PW: focused-seeds/geometry/S9-morph-ample-family-ample-relative/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $f : X \to Y$ be a morphism of schemes. Assume*

(1) *Y has an ample family of invertible modules,*
(2) *there exists an f-ample invertible module on X.*

Then X has an ample family of invertible modules.

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Proof. Let $\mathcal{L}$ be an f-ample invertible module on X. This in particular implies that f is quasi-compact, see Morphisms, Definition 01VH. Since Y is quasi-compact by Morphisms, Definition 0FXR we see that X is quasi-compact (and hence X itself satisfies the first condition of Morphisms, Definition 0FXR). Let $x \in X$ with image $y \in Y$. By assumption (2) we can find an invertible $\mathcal{O}_Y$-module $\mathcal{N}$ and a section $t \in \Gamma(Y, \mathcal{N})$ such that the locus Y_t where t does not vanish is affine. Then $\mathcal{L}$ is ample over $f^{-1}(Y_t) = X_{f^*t}$ and hence we can find a section $s \in \Gamma(X_{f^*t}, \mathcal{L})$ such that $(X_{f^*t})_s$ is affine and contains x. By Properties, Lemma 01PW for some $n \geq 0$ the product $(f^*t)^n s$ extends to a section $s' \in \Gamma(X, f^*\mathcal{N}^{\otimes n} \otimes \mathcal{L})$. Then finally the section $s'' = f^*ts'$ of $f^*\mathcal{N}^{\otimes n+1} \otimes \mathcal{L}$ vanishes at every point of $X \setminus X_{f^*t}$ hence we see that $X_{s''} = (X_{f^*t})_s$ is affine as desired. $\square$